

LOGISTIC REGRESSION

Homework #3

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1 Introduction

We build a binary logistic regression model on a data set containing information on crime for various neighborhoods of a major city to predict whether the neighborhood will be at risk for high crime levels. Each record in our data has a response variable indicating whether or not the crime rate is above the median crime rate or not. Our R statistical programming code is included in Appendix A.

2 Data Exploration

Note

Data exploration section completed.

The crime training data set contains 466 rows and 13 features, and is 100 percent complete. The data are all numeric, and the target variable is **target**, which represents whether the crime rate is above the median crime rate (1) or not (0). The other variables represent various information on crime for neighborhoods of a major city. See Appendix B for a full list of variable names and definitions. Table 1 presents descriptive characteristics of the training data.

Table 1: Summary statistics for all variables in the training dataset

Variable	Completeness rate	Mean	Min	P25	P50	P75	Max
zn	1	12.000	0.00	0.00	0.00	16.00	100.00
indus	1	11.000	0.46	5.10	9.70	18.00	28.00
chas	1	0.071	0.00	0.00	0.00	0.00	1.00

Table 1: Summary statistics for all variables in the training dataset

Variable	Completeness rate	Mean	Min	P25	P50	P75	Max
nox	1	0.550	0.39	0.45	0.54	0.62	0.87
rm	1	6.300	3.90	5.90	6.20	6.60	8.80
age	1	68.000	2.90	44.00	77.00	94.00	100.00
dis	1	3.800	1.10	2.10	3.20	5.20	12.00
rad	1	9.500	1.00	4.00	5.00	24.00	24.00
tax	1	410.000	190.00	280.00	330.00	670.00	710.00
ptratio	1	18.000	13.00	17.00	19.00	20.00	22.00
lstat	1	13.000	1.70	7.00	11.00	17.00	38.00
medv	1	23.000	5.00	17.00	21.00	25.00	50.00
target	1	0.490	0.00	0.00	0.00	1.00	1.00

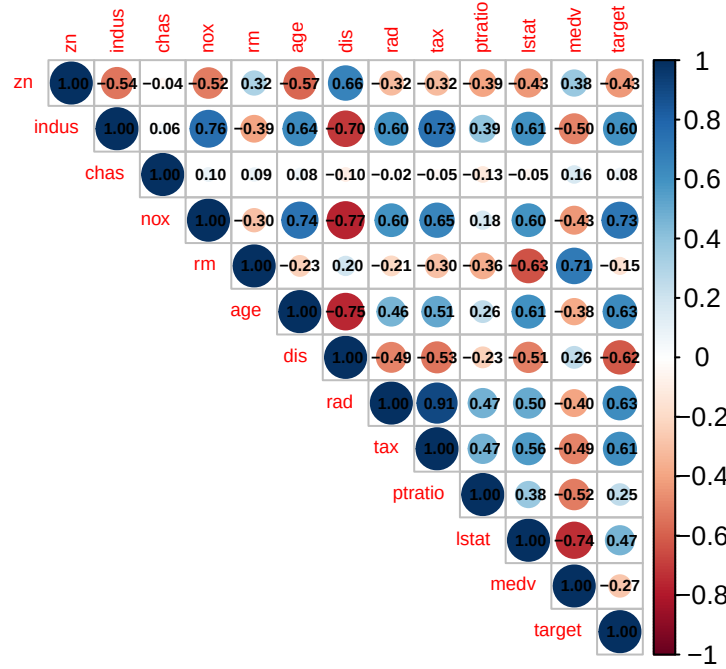


Figure 1: Correlation matrix of all variables in the training dataset

The correlation between all variables are shown Figure 1. The most correlated predictor variables with **target** are **nox** (nitrogen oxides concentration, $R = 0.73$), **age** (proportion of owner-occupied units built prior to 1940, $R = 0.63$), **rad** (index of accessibility to radial highways, $R = 0.63$), **dis** (weighted mean of distances to five Boston employment centers, $R = -0.62$), **tax** (full-value property-tax rate per \$10,000, $R = 0.61$), and **indus** (proportion of non-retail business acres per suburb, $R = 0.6$). These variables have the strongest linear relationships with the target variable and may be good predictors of crime rates.

Table 2: Multicollinear variables in the training dataset

Predictor 1	Predictor 2	Correlation
rad	tax	0.91
nox	dis	-0.77
indus	nox	0.76
age	dis	-0.75
lstat	medv	-0.74
nox	age	0.74
indus	tax	0.73
nox	target	0.73
rm	medv	0.71
indus	dis	-0.70

Table 2 shows predictor pairs with a pairwise correlation coefficient greater than 0.7. High pairwise correlation can create modeling issues because it becomes harder to isolate each predictor's individual effect on the target variable. The strongest multicollinear pair is **rad** (accessibility to highways) and **tax** (property-tax rate), which is plausible because neighborhoods with better highway access may also have higher property taxes. For this pair, and for the other pairs in the table, we should consider dropping one variable or using regularization techniques to mitigate multicollinearity.

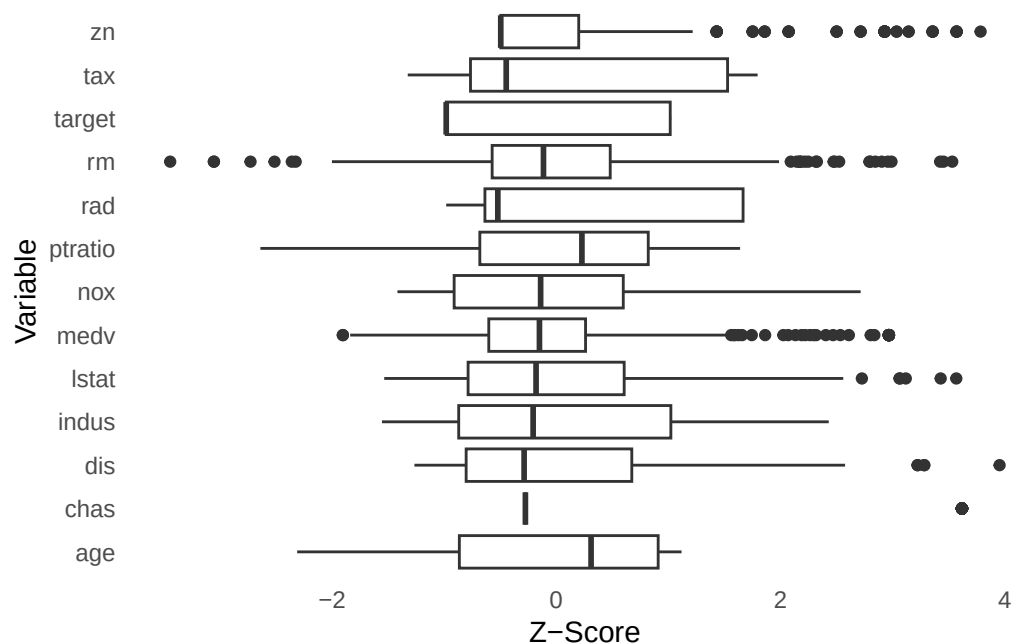


Figure 2: Box plots for identifying outliers in the training data

Figure 2 shows the distribution of z-scores for all variables and highlights potential outliers in the training data. The variable **zn** (proportion of residential land zoned for large lots over 25,000 square

feet) has many high outliers, which is plausible because some neighborhoods may have much higher shares of large lots than others; its maximum value is 100 percent. The variable `rm` (average number of rooms per dwelling) also has several outliers, ranging from 3.863 to 8.78 rooms. Likewise, `medv` (median value of owner-occupied homes) ranges from \$5k to \$50k, which is not unexpected given variation in housing prices across neighborhoods and unspecified data years. The variables `lstat` (percent lower status population) and `dis` (weighted mean distance to five Boston employment centers) show a few high outliers (37.97 percent and 12.1265 units, respectively). The variable `chas` has a high value of 1, which is expected because it is a binary indicator of whether a suburb borders the Charles River. Overall, these outliers appear to reflect real neighborhood variability rather than data-entry errors. If influence diagnostics later reveal problematic leverage points, we can consider adding indicator variables to account for them.

3 Data Preparation

Note

Completed

I don't think we need to compare this dataset to the baseball dataset (Homework 1).

For the crime data used in Homework 3, the first step in data preparation was to check whether any variables contained missing values. This crime data set does not contain missing values in either the training or evaluation files. Therefore, no imputation or missingness flag variables were needed.

Even though missing-value treatment was unnecessary, we still prepared the data for binary logistic regression by creating transformed and derived predictors. Several predictors are right-skewed, especially `zn`, `dis`, `rad`, `tax`, and `lstat`. For these variables, log transformations were created using `log1p()`, which reduces skewness while safely handling zero values. These transformed variables may improve model fit by making relationships with the log-odds of the response more stable.

We also created a small set of engineered variables that may be useful in later models. First, `nox_indus` captures the combined effect of air pollution concentration and industrial land use. Second, `rm_lstat_ratio` measures the balance between average rooms per dwelling and lower-status population, combining a housing-quality measure with a socioeconomic measure. Third, `tax_rad_ratio` captures the relationship between tax burden and highway accessibility. These new variables were created because logistic regression can benefit from transformed and combined predictors when the original variables may not fully capture the structure related to high-crime neighborhoods.

Finally, the cleaned training and evaluation data sets were stored as `crime_clean` and `crime_eval_clean` so the later modeling section can use them directly.

```
# Check missing values in training and evaluation data
missing_train <- crime_train |>
  summarise(across(everything(), ~ sum(is.na(.)))) |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Missing")

missing_eval <- crime_eval |>
  summarise(across(everything(), ~ sum(is.na(.)))) |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Missing")

missing_train
```

```
# A tibble: 13 x 2
  Variable Missing
  <chr>      <int>
1 zn          0
2 indus       0
3 chas        0
4 nox         0
5 rm          0
6 age         0
7 dis         0
8 rad         0
9 tax         0
10 ptratio    0
11 lstat      0
12 medv       0
13 target     0
```

```
missing_eval
```

```
# A tibble: 12 x 2
  Variable Missing
  <chr>      <int>
1 zn          0
2 indus       0
3 chas        0
4 nox         0
5 rm          0
6 age         0
```

7 dis	0
8 rad	0
9 tax	0
10 ptratio	0
11 lstat	0
12 medv	0

```
#| label: data-preparation-transformations
#| echo: true
#| warning: false

# Start with copies of the original datasets
crime_clean <- crime_train
crime_eval_clean <- crime_eval

# Create transformed and engineered predictors
crime_clean <- crime_clean |>
  mutate(
    # Log transforms for right-skewed predictors
    log_zn    = log1p(zn),
    log_dis   = log1p(dis),
    log_rad   = log1p(rad),
    log_tax   = log1p(tax),
    log_lstat = log1p(lstat),

    # Engineered variables
    nox_indus = nox * indus,
    rm_lstat_ratio = rm / lstat,
    tax_rad_ratio = tax / rad
  )

crime_eval_clean <- crime_eval_clean |>
  mutate(
    # Log transforms for right-skewed predictors
    log_zn    = log1p(zn),
    log_dis   = log1p(dis),
    log_rad   = log1p(rad),
    log_tax   = log1p(tax),
    log_lstat = log1p(lstat),
```

```
# Engineered variables
nox_indus      = nox * indus,
rm_lstat_ratio = rm / lstat,
tax_rad_ratio  = tax / rad
)

#| label: data-preparation-verify
#| echo: true

# Verify dimensions after preparation
dim(crime_clean)
```

```
[1] 466  21
```

```
dim(crime_eval_clean)
```

```
[1] 40 20
```

```
# Verify that no missing values remain after transformations
crime_clean |>
  summarise(across(everything(), ~ sum(is.na(.)))) |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Missing") |>
  filter(Missing > 0)
```

```
# A tibble: 0 x 2
# i 2 variables: Variable <chr>, Missing <int>
```

```
crime_eval_clean |>
  summarise(across(everything(), ~ sum(is.na(.)))) |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Missing") |>
  filter(Missing > 0)
```

```
# A tibble: 0 x 2
# i 2 variables: Variable <chr>, Missing <int>
```

```

### Linearity Assumption Summary

# The results above indicate that the assumption of linearity is satisfied for all independent

## Data Splits

# First, we will split our dataset into a training and testing set. We will use the training s

#| label: split-data
#| eval: false

set.seed(19940211)
train_indices <- sample(nrow(crime_clean), size = 0.8 * nrow(crime_clean))
crime_train_split <- crime_clean[train_indices, ]
crime_test_split <- crime_clean[-train_indices, ]

```

4 Check assumptions for binary logistic regression

1. Binary Outcome

```

# Confirming that the target is binary
crime_clean |>
  count(target) |>
  mutate(prop = n / sum(n))

```

```

# A tibble: 2 x 3
  target     n prop
  <dbl> <int> <dbl>
1      0   237 0.509
2      1   229 0.491

```

The response variable “target” is confirmed to be a binary, therefore it takes two values (0: crime rate below the median) and (1: crime rate above the median). In the 466 neighborhoods in training data set 237 (50.9%) fall in the class of low crime and 228 (49.1)% are in the high crime neighborhood. There is no class imbalance, due to the near equal split meaning there is unlikely to be bias towards one class.

2. Independence of Observations

Each record in the dataset represents a distinct neighborhood, and there is no repeated-measures or clustered structure in the data. Therefore, the independence of observations assumption is satisfied by design and requires no further adjustment.

3. No Multicollinearity (VIF Check)

```
library(car)

# Fit a reference logistic model with original predictors
model_ref <- glm(target ~ zn + indus + chas + nox + rm + age + dis + rad + tax + ptratio
                 + lstat + medv,
                 data = crime_clean, family = binomial)

# Compute VIF
vif_vals <- vif(model_ref) |>
  as.data.frame() |>
  rownames_to_column("Variable") |>
  rename(VIF = 2) |>
  arrange(desc(VIF))

vif_vals |> knitr::kable(caption = "Variance Inflation Factors for Reference Model")
```

Table 3: Variance Inflation Factors for Reference Model

Variable	VIF
medv	8.122037
rm	5.813851
nox	4.160497
dis	3.887981
indus	2.682271
lstat	2.642656
age	2.569961
ptratio	2.275557
tax	2.144040
rad	1.942967
zn	1.823146
chas	1.241479

The VIF values is calculated for all predictors to check for multicollinearity. A VIF above 10 is considered to be severe, while between 5 and 10 should warrant caution. None of the variable

exceed a threshold of 10, indicating multicollinearity is not present. The highest VIF belongs to medv (median home value, $VIF = 8.12$), followed by rm (average rooms per dwelling, $VIF = 5.81$). These two variables are moderate and do not need require any removal. The remaining predictors fall below 5 and are acceptable levels of multicollinearity. There was strong pairwise correlation between “rad” and “tax” in the exploration section, however it is confirmed that multicollinearity does not exist in the dataset.

4. Linearity of Log-Odds (Box-Tidwell Test)

```
# Box-Tidwell test: add interaction of each continuous predictor with its log
# to test whether the log-odds relationship is linear
continuous_vars <- c("zn", "indus", "nox", "rm", "age", "dis", "rad", "tax",
                    "ptratio", "lstat", "medv")

# Create interaction terms x * log(x) for each continuous predictor
# (use log1p to handle zeros)
crime_bt <- crime_clean |>
  mutate(across(all_of(continuous_vars), ~ . * log1p(.), .names = "{.col}_logx"))

bt_formula <- as.formula(paste(
  "target ~",
  paste(continuous_vars, collapse = " + "),
  "+",
  paste(paste0(continuous_vars, "_logx"), collapse = " + ")
))

model_bt <- glm(bt_formula, data = crime_bt, family = binomial)

# Extract p-values for the interaction terms only
bt_results <- summary(model_bt)$coefficients |>
  as.data.frame() |>
  rownames_to_column("Term") |>
  filter(str_detect(Term, "_logx")) |>
  mutate(Variable = str_remove(Term, "_logx"),
         Significant = `Pr(>|z|)` < 0.05) |>
  select(Variable, Estimate, `Pr(>|z|)`, Significant) |>
  mutate(across(where(is.numeric), ~ signif(., 3)))

bt_results |> knitr::kable(caption = "Box-Tidwell Test for Linearity of Log-Odds")
```

Table 4: Box-Tidwell Test for Linearity of Log-Odds

Variable	Estimate	Pr(> z)	Significant
zn	-0.0387	0.803000	FALSE
indus	0.2700	0.390000	FALSE
nox	328.0000	0.128000	FALSE
rm	31.6000	0.003110	TRUE
age	0.0699	0.163000	FALSE
dis	-4.9100	0.000615	TRUE
rad	0.7500	0.074900	FALSE
tax	-0.2380	0.000608	TRUE
ptratio	4.7200	0.090500	FALSE
lstat	0.1810	0.382000	FALSE
medv	-0.1540	0.616000	FALSE

The Box-Tidwell test was used to assess whether each continuous predictor has a linear relationship with the log-odds of the outcome. A p-value greater than 0.05 violate the assumption. There are three variables that were flagged as significant:rm ($p = 0.003$), dis ($p < 0.001$), and tax ($p < 0.001$). This indicates that these predictors do not have strict linear relationship with log-odds of high crime. Log transformation were applied to these variables in the data preparation section in order to be used in the model building. The remaining eight predictors did not have any evidence of violating the linearity assumption.

5 Build models

Note

Completed but file won't render. The rendering fails on line 265, but I'm not sure what's wrong. Any ideas?

We compared three models. Our first model included all the variables in the cleaned training dataset. The second model aimed to reduce multicollinearity by omitting variables that had correlation coefficients $|r| > 0.7$ (see Data Exploration section). Finally, we generated the third model using backward stepwise selection from a full model that included transformed variables and newly created variables (see Data Preparation section).

5.1 Model 1. All untransformed variables

As shown below, the most significant predictors were `nox` and `rad` ($P < 0.001$). In addition, `age` and `ptratio` were significant at the $P < 0.01$ level, and `dis` and `tax` were significant at the $P < 0.05$ level. The AIC for this model is `r\ AIC(model1)`.

```
model1 <- glm(
  formula = target ~ zn + indus + chas + nox + rm + age + dis + rad + tax +
    ptratio + lstat + medv,
  family = "binomial",
  data = crime_train_split)

summary(model1)
```

Call:

```
glm(formula = target ~ zn + indus + chas + nox + rm + age + dis +
  rad + tax + ptratio + lstat + medv, family = "binomial",
  data = crime_train_split)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-47.929725	8.808610	-5.441	5.29e-08	***
zn	-0.048895	0.037182	-1.315	0.18851	
indus	-0.054428	0.066155	-0.823	0.41065	
chas	0.924573	1.061844	0.871	0.38390	
nox	55.009200	10.291959	5.345	9.05e-08	***
rm	0.050251	0.882349	0.057	0.95458	
age	0.048050	0.017874	2.688	0.00718	**
dis	0.657857	0.260376	2.527	0.01152	*
rad	0.783482	0.201429	3.890	0.00010	***
tax	-0.010043	0.004025	-2.495	0.01259	*
ptratio	0.475152	0.159502	2.979	0.00289	**
lstat	0.010310	0.069436	0.148	0.88196	
medv	0.127463	0.078800	1.618	0.10576	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 515.31  on 371  degrees of freedom
Residual deviance: 133.42  on 359  degrees of freedom
AIC: 159.42

```

```
Number of Fisher Scoring iterations: 9
```

5.2 Model 2. Reduce multicollinearity

In the Data Exploration section, we showed that **rad** and **tax** were strongly correlated ($r=0.91$), which suggests that only one of these variables is needed in Model 1 (we chose to retain **rad** as it was more highly significant). Similarly, **medv** was shown to be potentially multicollinear with **lstat** ($r=0.74$) and **rm** ($r=0.71$), so we removed the latter variables.

In addition, the correlation between **nox** (predictor) and **target** (response) suggests that **nox** is an important predictor, so we removed its potential collinear pairs **age** and **dis**. Although **nox** was moderately correlated with **indus** ($r=0.76$), we retained **indus** because removal of both **indus** and **tax** (highly correlated with **rad**) would eliminate a potentially informative predictor.

These changes resulted in a model with AIC $r \setminus$ AIC(model2).

```

model2 <- glm(
  formula = target ~ zn + indus + chas + nox + rad + ptratio + medv,
  family = "binomial",
  data = crime_train_split)

summary(model2)

```

Call:

```

glm(formula = target ~ zn + indus + chas + nox + rad + ptratio +
     medv, family = "binomial", data = crime_train_split)

```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-34.11942	6.07656	-5.615	1.97e-08	***
zn	-0.03024	0.02862	-1.057	0.290641	
indus	-0.15005	0.05644	-2.658	0.007851	**
chas	1.45828	0.89598	1.628	0.103614	
nox	45.57838	8.64114	5.275	1.33e-07	***
rad	0.46600	0.14094	3.306	0.000946	***
ptratio	0.35450	0.13212	2.683	0.007291	**

```
medv          0.08292    0.03159    2.625 0.008656 **
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for binomial family taken to be 1)
```

```
Null deviance: 515.31  on 371  degrees of freedom
```

```
Residual deviance: 157.41  on 364  degrees of freedom
```

```
AIC: 173.41
```

```
Number of Fisher Scoring iterations: 9
```

5.3 Model 3. Backward stepwise variable selection

For this model, we performed backward stepwise selection from the set of variables including transformed and newly created variables.

The best model had `nox`, `tax_rad_ratio`, `ptratio`, and `age` as the most significant predictors ($P < 0.001$). In addition, `log_dis` and `rm_lstat_ratio` were significant at the $P < 0.05$ level. The AIC of this model was `r\ AIC(model3)`, which was lower than that of Models 1 and 2.

```
# Full model includes all variables
model_full <- glm(
  formula = target ~ log_zn + indus + chas + nox + rm + age +
    log_dis + log_rad + log_tax + ptratio + log_lstat + medv +
    nox_indus + rm_lstat_ratio + tax_rad_ratio,
  family = "binomial",
  data = crime_train_split)

# Select variables
model3 <- step(model_full, direction = "backward")
```

```
Start:  AIC=158.32
```

```
target ~ log_zn + indus + chas + nox + rm + age + log_dis + log_rad +
  log_tax + ptratio + log_lstat + medv + nox_indus + rm_lstat_ratio +
  tax_rad_ratio
```

```
Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
Df Deviance    AIC
```

- log_tax	1	126.46	156.46
- rm	1	126.47	156.47
- log_zn	1	126.69	156.69
- log_rad	1	127.06	157.06
- log_lstat	1	127.16	157.16
- tax_rad_ratio	1	127.38	157.38
- medv	1	127.45	157.45
- indus	1	127.91	157.91
- chas	1	127.96	157.96
- nox_indus	1	128.02	158.02
<none>		126.32	158.32
- rm_lstat_ratio	1	131.00	161.00
- log_dis	1	132.94	162.94
- age	1	136.32	166.32
- ptratio	1	136.95	166.95
- nox	1	162.12	192.12

Step: AIC=156.46

target ~ log_zn + indus + chas + nox + rm + age + log_dis + log_rad +
 ptratio + log_lstat + medv + nox_indus + rm_lstat_ratio +
 tax_rad_ratio

	Df	Deviance	AIC
- rm	1	126.62	154.62
- log_zn	1	126.82	154.82
- log_lstat	1	127.23	155.23
- medv	1	127.73	155.73
- indus	1	127.92	155.92
- nox_indus	1	128.05	156.05
- log_rad	1	128.31	156.31
- chas	1	128.43	156.43
<none>		126.46	156.46
- rm_lstat_ratio	1	131.03	159.03
- tax_rad_ratio	1	131.18	159.18
- log_dis	1	133.51	161.51
- age	1	136.32	164.32
- ptratio	1	137.07	165.07
- nox	1	166.00	194.00

Step: AIC=154.62

```
target ~ log_zn + indus + chas + nox + age + log_dis + log_rad +
      ptratio + log_lstat + medv + nox_indus + rm_lstat_ratio +
      tax_rad_ratio
```

	Df	Deviance	AIC
- log_zn	1	127.11	153.11
- log_lstat	1	127.57	153.57
- indus	1	128.03	154.03
- medv	1	128.13	154.13
- nox_indus	1	128.16	154.16
- log_rad	1	128.31	154.31
- chas	1	128.55	154.55
<none>		126.62	154.62
- rm_lstat_ratio	1	131.04	157.04
- tax_rad_ratio	1	131.58	157.58
- log_dis	1	133.53	159.53
- ptratio	1	137.47	163.47
- age	1	137.57	163.57
- nox	1	166.90	192.90

Step: AIC=153.11

```
target ~ indus + chas + nox + age + log_dis + log_rad + ptratio +
      log_lstat + medv + nox_indus + rm_lstat_ratio + tax_rad_ratio
```

	Df	Deviance	AIC
- log_lstat	1	127.78	151.78
- medv	1	128.47	152.47
- log_rad	1	128.60	152.60
- indus	1	128.94	152.94
- nox_indus	1	129.06	153.06
<none>		127.11	153.11
- chas	1	129.54	153.54
- rm_lstat_ratio	1	131.31	155.31
- tax_rad_ratio	1	133.10	157.10
- log_dis	1	133.55	157.55
- age	1	138.77	162.77
- ptratio	1	144.07	168.07
- nox	1	170.11	194.11

Step: AIC=151.78

```
target ~ indus + chas + nox + age + log_dis + log_rad + ptratio +
      medv + nox_indus + rm_lstat_ratio + tax_rad_ratio
```

	Df	Deviance	AIC
- medv	1	129.14	151.14
- log_rad	1	129.55	151.55
- indus	1	129.69	151.69
<none>		127.78	151.78
- nox_indus	1	129.80	151.80
- chas	1	130.22	152.22
- rm_lstat_ratio	1	132.54	154.54
- tax_rad_ratio	1	133.46	155.46
- log_dis	1	135.48	157.48
- age	1	143.24	165.24
- ptratio	1	144.43	166.43
- nox	1	171.35	193.35

Step: AIC=151.14

```
target ~ indus + chas + nox + age + log_dis + log_rad + ptratio +
      nox_indus + rm_lstat_ratio + tax_rad_ratio
```

	Df	Deviance	AIC
- log_rad	1	130.47	150.47
- indus	1	130.74	150.74
- nox_indus	1	130.86	150.86
<none>		129.14	151.14
- chas	1	131.81	151.81
- log_dis	1	135.48	155.48
- tax_rad_ratio	1	136.34	156.34
- age	1	143.85	163.85
- ptratio	1	145.01	165.01
- rm_lstat_ratio	1	150.57	170.57
- nox	1	171.57	191.57

Step: AIC=150.47

```
target ~ indus + chas + nox + age + log_dis + ptratio + nox_indus +
      rm_lstat_ratio + tax_rad_ratio
```

	Df	Deviance	AIC
- indus	1	132.01	150.01
- nox_indus	1	132.02	150.02
<none>		130.47	150.47
- chas	1	133.15	151.15
- log_dis	1	135.90	153.90
- age	1	145.28	163.28
- ptratio	1	147.07	165.07
- rm_lstat_ratio	1	151.57	169.57
- nox	1	172.02	190.02
- tax_rad_ratio	1	177.44	195.44

Step: AIC=150.01

target ~ chas + nox + age + log_dis + ptratio + nox_indus + rm_lstat_ratio +
tax_rad_ratio

	Df	Deviance	AIC
- nox_indus	1	132.02	148.02
<none>		132.01	150.01
- chas	1	134.98	150.98
- log_dis	1	137.55	153.55
- age	1	146.75	162.75
- ptratio	1	148.26	164.26
- rm_lstat_ratio	1	151.61	167.61
- tax_rad_ratio	1	184.46	200.46
- nox	1	186.02	202.02

Step: AIC=148.02

target ~ chas + nox + age + log_dis + ptratio + rm_lstat_ratio +
tax_rad_ratio

	Df	Deviance	AIC
<none>		132.02	148.02
- chas	1	135.02	149.02
- log_dis	1	137.66	151.66
- age	1	147.04	161.04
- ptratio	1	148.29	162.29
- rm_lstat_ratio	1	151.61	165.61

```
- tax_rad_ratio    1    189.81 203.81
- nox              1    215.50 229.50
```

```
summary(model3)
```

Call:

```
glm(formula = target ~ chas + nox + age + log_dis + ptratio +
     rm_lstat_ratio + tax_rad_ratio, family = "binomial", data = crime_train_split)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-44.89858	7.47464	-6.007	1.89e-09	***
chas	1.55357	0.90952	1.708	0.087613	.
nox	55.23868	8.90173	6.205	5.46e-10	***
age	0.05359	0.01503	3.565	0.000364	***
log_dis	2.64074	1.12460	2.348	0.018867	*
ptratio	0.50438	0.13160	3.833	0.000127	***
rm_lstat_ratio	2.52680	0.65649	3.849	0.000119	***
tax_rad_ratio	-0.05751	0.01302	-4.418	9.95e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 515.31 on 371 degrees of freedom
 Residual deviance: 132.02 on 364 degrees of freedom
 AIC: 148.02

Number of Fisher Scoring iterations: 8

Of these three models, model 3 has the lowest AIC. In the next section, we will evaluate the performance of these models to more determine which one is best for predicting the `target` variable.

6 Select Models

```

# Creating a custom helper function to extract all the required metrics
# explicitly evaluating on the provided training dataset.
get_logistic_metrics <- function(model, model_name, data, target_var = "target") {
  # 1. Get predicted probabilities and classes
  probs <- predict(model, newdata = data, type = "response")
  preds <- factor(ifelse(probs > 0.5, 1, 0), levels = c(0, 1))
  actuals <- factor(data[[target_var]], levels = c(0, 1))

  # 2. Generate Confusion Matrix and derived metrics
  cm <- confusionMatrix(preds, actuals, positive = "1")

  # 3. Calculate AUC
  roc_obj <- roc(data[[target_var]], probs, quiet = TRUE)
  auc_val <- as.numeric(pROC::auc(roc_obj))

  # 4. Return results as a tibble
  tibble(
    Model = model_name,
    Deviance = model$deviance,
    Accuracy = cm$overall["Accuracy"],
    Error_Rate = 1 - cm$overall["Accuracy"],
    Precision = cm$byClass["Precision"],
    Sensitivity = cm$byClass["Sensitivity"], # Also Recall
    Specificity = cm$byClass["Specificity"],
    F1 = cm$byClass["F1"],
    AUC = auc_val,
    AIC = AIC(model),
    BIC = BIC(model)
  )
}

models_list <- list(
  "All Untransformed Variables" = model1,
  "Reducing Multicollinearity" = model2,
  "Full Model with Backward Selection" = model_full
)

# Apply the function to all three models, explicitly passing the training split
model_comparison <- imap_dfr(models_list, ~{

```

```

  get_logistic_metrics(.x, .y, crime_test_split)
}) |>
  arrange(`Deviance`)

# Transpose the data and convert to a data frame
model_t <- as.data.frame(t(model_comparison))

# 3. Display the table
knitr::kable(
  model_t,
  digits = 4,
  caption = "Evaluation of Candidate Models on Training Dataset"
)

```

Table 5: Evaluation of Candidate Models on Training Dataset

	V1	V2	V3
Model	Full Model with Backward Selection	All Untransformed Variables	Reducing Multicollinearity
Deviance	126.3233	133.4153	157.4125
Accuracy	0.8829787	0.8404255	0.8617021
Error_Rate	0.1170213	0.1595745	0.1382979
Precision	0.9318182	0.8541667	0.9090909
Sensitivity	0.8367347	0.8367347	0.8163265
Specificity	0.9333333	0.8444444	0.9111111
F1	0.8817204	0.8453608	0.8602151
AUC	0.9414966	0.9410431	0.9414966
AIC	158.3233	159.4153	173.4125
BIC	221.0256	210.3609	204.7636

```

best_model_info <- model_comparison |> slice(1)

best_model_obj <- models_list[[best_model_info$Model]]

```

For the logistic models, we decided to use the deviance as our primary criterion on the training dataset, which was attained by the Full Model with Backward Selection. We've chosen deviance as our primary metric because it directly measures the goodness of fit for logistic regression models,

with lower deviance indicating a better fit to the training data. Given our model results, each model had a fairly similar accuracy with our Full Model with Backward Selection having the highest accuracy of 0.883. It also had the lowest error rate of 0.117 and was comparable with the other top models in sensitivity and specificity. Overall, this model had the best balance of fit and predictive performance on the training data, which is why we selected it for further evaluation on the test set and for generating predictions on the evaluation set.

Given that our target variable was fairly balanced (49% of all observations had a target value of 1), and our accuracy, Precision, and Sensitivity show that we greatly outperform a random classifier, we are confident that our model is performing well.

```
# Display the summary of the selected Stepwise model to interpret coefficients and significance
summary(best_model_obj)
```

To make inferences from our top performing model, we can look at the significance of the predictors. The logistic regression model identifies several key environmental and structural factors that significantly predict whether a suburb's crime rate will exceed the median. The most dominant predictor is the concentration of nitrogen oxides (nox), where higher levels are strongly associated with a higher probability of the area being high-crime ($z=4.670, p<0.001$). Additionally, the pupil-teacher ratio (ptratio) and the proportion of older homes (age) built before 1940 are both significant positive predictors, suggesting that areas with crowded schools and aging infrastructure are more likely to see elevated crime rates. The model also highlights that as the distance to employment centers (log_dis) increases, so does the likelihood of high crime, along with a significant effect from the ratio of rooms to lower-status population (rm_lstat_ratio). Interestingly, many traditional economic indicators, such as median home values (medv), property tax rates (log_tax), and zoning for large lots (log_zn), did not reach statistical significance, indicating that in this specific dataset, industrial and demographic markers are more reliable indicators of crime than general wealth or proximity to the Charles River.

7 Results

```
# Generate predictions on the evaluation dataset using the selected model
eval_predictions <- predict(best_model_obj, newdata = crime_eval_clean, type = "response")

crime_eval_results <- crime_eval_clean |>
  mutate(
    # If probability > 0.5, it's a 1 (high crime), otherwise 0
    PREDICTED_TARGET = ifelse(eval_predictions > 0.5, 1, 0)
  )
```

```
# Write predictions to a csv
write_csv(crime_eval_results, "crime_evaluation_predictions.csv")

# Display the first 10 predictions in the rendered PDF
knitr::kable(
  head(crime_eval_results |> select(PREDICTED_TARGET), 10),
  caption = "Excerpts of Stepwise Model Win Predictions for Evaluation Set"
)
```

Given our evaluation set and our stepwise model, we are able to create predictions on the evaluation set on which neighborhoods are classified as high-crime or low-crime. The first 10 predictions are shown in the table above. We write these predictions to a CSV file for submission.

8 Conclusion

In conclusion, our Full Model with Backward Selection model, which was selected based on deviance and other performance metrics, demonstrates strong predictive capabilities for classifying neighborhoods as high-crime or low-crime based on a variety of environmental and structural factors. The significant predictors identified in the model provide valuable insights into the underlying drivers of crime in urban areas, highlighting the importance of air pollution, school crowding, aging infrastructure, and proximity to employment centers. These findings can inform policymakers and community leaders in their efforts to target interventions and allocate resources effectively to reduce crime rates. Future work could explore additional modeling techniques, such as regularization or ensemble methods, to further enhance predictive performance and interpretability.

9 Appendix A: R Code

Note

If we want to include our R statistical programming code in an Appendix, add all code here and set the chunk option `include = false` to hide the code from the main report.

10 Appendix B: Data Dictionary

Variable	Description	Role
zn	Proportion of residential land zoned for large lots (over 25,000 square feet)	Predictor
indus	Proportion of non-retail business acres per suburb	Predictor
chas	Dummy variable for whether the suburb borders the Charles River (1) or not (0)	Predictor
nox	Nitrogen oxides concentration (parts per 10 million)	Predictor
rm	Average number of rooms per dwelling	Predictor
age	Proportion of owner-occupied units built prior to 1940	Predictor
dis	Weighted mean of distances to five Boston employment centers	Predictor
rad	Index of accessibility to radial highways	Predictor
tax	Full-value property-tax rate per \$10,000	Predictor
ptratio	Pupil-teacher ratio by town	Predictor
lstat	Lower status of the population (percent)	Predictor
medv	Median value of owner-occupied homes in \$1000s	Predictor
target	Whether the crime rate is above the median crime rate (1) or not (0)	Response